

ECON 101—Introduction to Microeconomics
Spring 2023

Midterm I

February 21, 2023

Name:

NU ID number:

Instructions:

- Answer all questions. Multiple choice answers go in the space provided below. Write your answers for the open problems in the space provided. You can use as many scratch paper for your preliminary calculations, but that work will not be considered. In the exam booklet write only your final arguments and answers.
 - The exam has a duration of 1 hour and 15 minutes, from 3 PM to 4:15 PM.
 - You are not allowed to go out during the duration of the exam. This means no bathroom breaks. Thus, be prepared.
 - You are only allowed to have in your desk area: ID, writing instruments, ruler, and a simple calculator (no graphing calculators), and a simple watch if needed (no smartwatches).
 - You must leave all other personal items, including smart devices, to the side of the room. It may be a better idea to not bring them at all.
- 5
- If you finish early the exam, you will only be allowed to go out until 3:50. Otherwise, you will have to remain in the room quietly until the exam is finished.
 - You are not allowed to remove the staple from your exam booklet.

Multiple choice. Please write your answers here!

1	2	3	4	5	6	7	8	9	10
a	c	c	b	a	d	c	b	a	b

Do not write below this area!

MCQ	Problem 1	Problem 2	Problem 3	Total

Part I: Multiple Choice

Please record all answers in the front sheet. Each question is worth 2 points for a total of 20 points.

Read carefully the following text and answer questions 1–5.

The economy of Kazakhstan produces the following five consumption goods: Food (F), Books (B), Entertainment (E), Clothing (C), and drugs (D).

The following conditions hold on the income elasticities of demand:

$$\epsilon_I^C < 0 < \epsilon_I^F < \epsilon_I^B < \epsilon_I^D < 1 < \epsilon_I^E.$$

Furthermore, we know that the following relationships are true:

- F and B are complementary.
- F and E are complementary.
- F and C are substitutes.
- F and D are substitutes.
- B and E are substitutes.
- B and D are complementary.
- B and C are complementary.
- E and C are complementary.
- E and D are complementary.

The supply curves of all the goods have positive slope.

Answer the following five questions, 1–2, in strict order, only taking into account the effects of the previous questions to your answer.

Suppose that there is a technological change in the production of books.

1. In the book (B) market:

- (a) The supply curve will shift to the right and the quantity demanded will increase.
- (b) The demand curve will shift to the right and the quantity offered will increase.
- (c) The demand and supply curves will shift.
- (d) Only the supply curve will shift to the right, neither the demand curve nor the quantity demanded will be affected.

2. In the entertainment (E) market (consider exclusively the change in price in the book market)

- (a) The price and the equilibrium quantity will increase and the producer surplus will increase.
- (b) The price and the equilibrium quantity will decrease and the consumer surplus will increase.
- (c) The price and the equilibrium quantity will decrease and the producer surplus will decrease.
- (d) The price will increase and the equilibrium quantity will decrease and the producer surplus will decrease.

3. In the food (F) market (consider exclusively the change in price in the book market and the entertainment market)

- (a) The demand curve will shift to the left and the quantity supplied will decrease.
- (b) The supply curve will shift to the right and the quantity demanded will increase.
- (c) The demand curve will shift to the right and the quantity supplied will increase.
- (d) The demand and supply curve will shift to the right.

4. In the drug (D) market (consider exclusively the change in price in the book, entertainment, and food markets)
 - (a) The demand curve will shift to the left and the equilibrium quantity will decrease.
 - (b) The demand curve will shift to the right and the equilibrium quantity will increase.
 - (c) The demand and supply curves will shift to the right.
 - (d) The demand curve will shift to the left exclusively, neither the supply curve nor the equilibrium quantity will be affected.

5. If the income of the population increases (consider exclusively the change in income)
 - (a) The demand curve of all goods except clothing will shift to the right.
 - (b) The demand curve for all goods will shift to the right.
 - (c) The demand curve for clothing and for entertainment will shift to the left.
 - (d) The price and the equilibrium quantity in all markets will increase.

6. Nurdaulet buys 4 pairs of jeans when the price is \$25. He buys 3 pairs when the price increases to \$50. Ryan's percentage change in quantity demanded is _____, when the price increased by _____ percent, and thus his price elasticity of demand is _____.
 - (a) -15% ; $+75\%$; -0.25
 - (b) $+25\%$; -100% ; -4
 - (c) -25% ; $+100\%$; -4
 - (d) -25% ; $+100\%$; -0.25

7. Suppose that the labor market is initially in equilibrium (L_1, w_1) . Which of the following *could not* cause an increase in the equilibrium wage rate from w_1 to w_2 .
 - (a) Government subsidy to firms who hire workers in this industry.
 - (b) An increase in the productivity of labor.
 - (c) A wage of immigration workers.
 - (d) An increase in the demand for the final product.

8. Flights to Paris are a normal good and people's incomes rise. At the same time, the price of jet fuel rises. The equilibrium price of a flight to Paris _____ and the equilibrium quantity of flights to Paris _____.
 - (a) rises; increases
 - (b) rises; might increase, decrease, or not change
 - (c) might rise, fall, or not change; increases
 - (d) falls; decreases.

9. Last year, one of the professors that teaches ECON101 went to the *Queen Tribute Concert* in the Astana Congress Center. From afar he observed a large number of desperate *resellers* at the entrance of the venue. His surprise was great when he approached some of them and asked for the reselling price of the tickets and noticed that the price at which they offered the ticket at that moment was less than what he paid in ticketon.kz a few weeks before the concert. An explanation of this phenomenon can be attributed to:
- (a) The demand for the *Queen Tribute concert* was lower than expected, therefore, at the selling price there was an excess supply of tickets and consequently that caused the reselling price of the tickets to decrease.
 - (b) The demand for the *Queen Tribute Concert* was higher than the expected demand, therefore, at the selling price there was an excess demand of tickets and consequently that caused the reselling price of the tickets to decrease.
 - (c) Reselling is illegal, thus, resellers were trying to get rid of the tickets before the police could catch them.
 - (d) The supply curve for tickets shifted to the left, therefore, there was an excess demand for tickets and consequently that caused the reselling price of the tickets to decrease.
10. The economic problem is the result of
- (a) Needs and wants of society are unlimited.
 - (b) Resources are limited in relation to needs and wants that are multiple.
 - (c) The resources available to society are abundant in relation to society's needs and wants.
 - (d) Resources are limited.

End of Multiple Choice Section: Please don't forget to record your answers in the answer sheet!

Continue to Part II

Part II: Long Problems

Please answer *only* in the space provided. Be neat and give complete arguments. You can use as much scratch paper as you need but your scratch work will not be considered.

Problem 1

A market analysis show that there are two types of consumers for coffee at Nazarbayev University. The first type are mild consumers of coffee and have an inverse demand curve of $P = 900 - 90Q$. The second type have a *caffeine addiction* and have an inverse demand curve of $P = 1200 - 60Q$. Quantities are in cups of coffee and the price is in tenge. In Nazarbayev University there are N_1 individuals who are mild consumers, and N_2 consumers with a *caffeine addiction*, where:

- $N_1 = 1000$
- $N_2 = 1500$

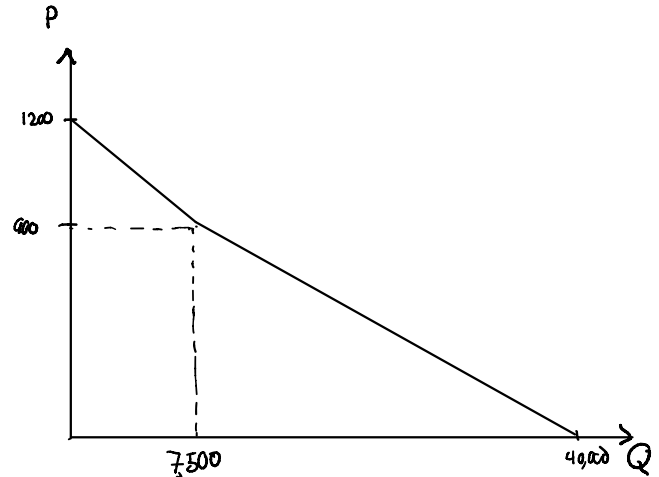
(a) (5 points) Using the demand curves above, derive the market demand for coffee in Nazarbayev University. Plot the market demand curve.

Type 1 demand: $q_1 = 10 - \frac{P}{90}$

Type 2 demand: $q_2 = 20 - \frac{P}{60}$

$$\begin{aligned} & 10,000 - \frac{100}{9}P \\ & + 30,000 - 25P \\ \hline & 40,000 - \frac{325}{9}P \end{aligned} \quad \text{Thus } Q_d(P) = \begin{cases} 40,000 - \frac{325}{9}P & \text{if } 0 \leq P \leq 900 \\ 30,000 - 25P & \text{if } 900 < P \leq 1200 \\ 0 & \text{if } P > 1200 \end{cases}$$

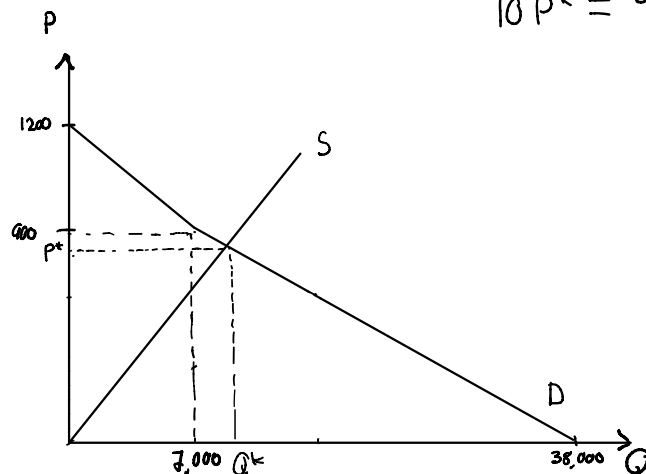
$$\begin{aligned} N_1 &= 1000 \text{ identical Type 1 consumers} \\ \Rightarrow Q_1 &= 1000 \left(10 - \frac{P}{90} \right) = 10,000 - \frac{100}{9}P \\ N_2 &= 1500 \text{ identical Type 2 consumers} \\ \Rightarrow Q_2 &= 1500 \left(20 - \frac{P}{60} \right) = 30,000 - 25P \\ Q_1 &\geq 0 \Leftrightarrow 10 - \frac{P}{90} \geq 0 \Leftrightarrow 900 \geq P \\ Q_2 &\geq 0 \Leftrightarrow 20 - \frac{P}{60} \geq 0 \Leftrightarrow 1200 \geq P \\ \therefore \text{aggregate demand } Q_d(P) &= \min\{Q_1(P), 0\} + \min\{Q_2(P), 0\} \\ \text{will have parts defined by } & 0 \leq P \leq 900, 900 < P \leq 1200, P > 1200 \end{aligned}$$



(b) (10 points) Suppose that the market supply for coffee in Nazarbayev University is given by $Q^S = 10P$. Find the equilibrium price and quantity. How much does each individual consumer buy at the equilibrium price?
Hint: Check the equilibrium price and quantity with your graph.

$$10P^* = 40,000 - \frac{325}{9}P^* \Leftrightarrow P^* = \frac{72,000}{83} \approx 867.47$$

$$Q^* = \frac{720,000}{83} \approx 8674.7$$



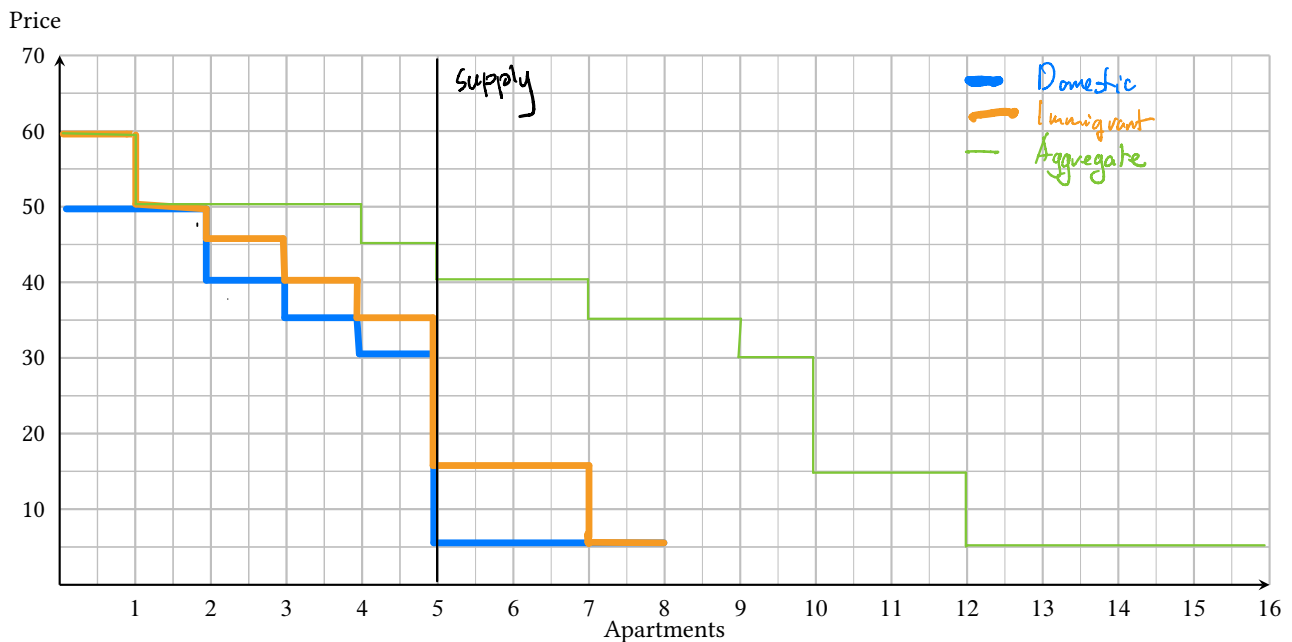
$$\begin{aligned} \text{Type 1 consumers buy: } & 10 - \frac{(72,000/83)}{90} = \frac{30}{83} \approx 0.3614 \\ \text{Type 2 consumers buy: } & 20 - \frac{(72,000/83)}{60} = \frac{460}{83} \approx 5.54 \end{aligned}$$

Problem 2

The reservation value or reservation price of a consumer is the price at which she is just indifferent between renting or not renting an apartment. At any price below the reservation price the consumer will demand one apartment, at any price above the reservation price the consumer will demand zero apartments, and exactly at the reservation price the consumer will be indifferent between having zero or one apartment.

In this problem you will consider one tiny part of the market for apartments in Astana. Due to the situation with Russia, the Astana housing market recently experienced an influx of Russian immigrants seeking to find housing within the city. Suppose that we have 8 Kazakh nationals and 8 Russian immigrants who want to rent an apartment. Their reservation prices are given below. Kazakh nationals are labeled with uppercase letters and the Russian immigrants with lowercase letters.

Person	A	B	C	D	E	F	G	H	a	b	c	d	e	f	g	h
Price	50	35	50	5	30	40	5	5	60	35	50	15	40	45	5	15



- (a) (5 points) Plot the domestic demand and immigrant demand curves in the graph above. Make sure to distinguish them by labeling them properly. *Hint: The demands will have a "staircase" shape. Arrange the numbers and identities first by decreasing reservation price and then alphabetically.*

A C F B E D G H a c f b e d g h
50 50 40 35 30 5 5 5 60 50 45 40 35 15 15 5

- (b) (5 points) What is the main difference that you see between the domestic and immigrant demand curves?

The immigrant demand curve has reservation values greater or equal than the reservation values of the local demand curve.

- (c) (5 points) Plot the aggregate demand for apartments in Astana in the graph above. Label it properly to distinguish it from the other two. *Hint: This is really no different from what you have done above. Arrange all numbers first by decreasing reservation price, then by nationality (first Kazakh then Russian), and finally alphabetically.*

a A C c f F e B b E d h D G H g
60 50 50 50 45 40 40 35 35 30 15 15 5 5 5 5

- (d) (5 points) Consider only the domestic demand. Suppose that if consumers are indifferent between renting or not, i.e., when their reservation value equals the rental price, they choose to rent. Suppose that the supply of apartments is fixed at 5 units. What is the highest price that would make the domestic demand for apartments equal to 5 units?

The highest price would be 30.

What is the lowest price that would make the domestic demand for apartments equal to 5 units?

The lowest price would be 5.

- (e) (5 points) Suppose that apartment owners have all the bargaining power and can charge the highest possible market price. Consider the domestic market only. Which of the A-H people end up getting apartments? What is their consumer surplus?

Persons: A C F B E

$$CS = 55 = (50 - 30) + (50 - 30) + (40 - 30) + (35 - 30)$$

- (f) (5 points) Now consider the aggregate demand. Continue assuming that if consumers are indifferent between renting or not, they choose to rent. Suppose, as in (c), that if renters of different nationalities have the same reservation price, the Kazakh one comes first. What is the highest price that would make the aggregate demand for apartments equal to 5 units?

The highest price would be 45.

What is the lowest price that would make the aggregate demand for apartments equal to 5 units?

The lowest price would be 40.

- (g) (5 points) Again, suppose that apartment owners can charge the highest possible market price. Which people among both locals and immigrants end up getting apartments?

Persons: a A C c f

- (h) (5 points) Consider the highest possible market price you found in (f) and the identifies of the *Astana* buyers you found in (g). What is the domestic consumer surplus?

$$CS = 10 = (50 - 45) + (50 - 45)$$

- (i) (5 points) Consider the aggregate demand and domestic consumer surplus. In order to help *Astana* renters, would you recommend putting a rent ceiling equal to the highest price you found in (d) and (e)?

The rent ceiling would be 30

→ Such a rent ceiling would help *Astana* renters A & C
making their surplus = $2(50 - 30) = 40$ $\Delta CS = 40 - 10 = 30$.

→ If we only consider *Astana* renters then this policy helps.

→ We would also need to consider local apartment owners but this question is not asking about them.

Problem 3

Kazakhstan's market for milk is described by the following domestic demand and supply schedules:

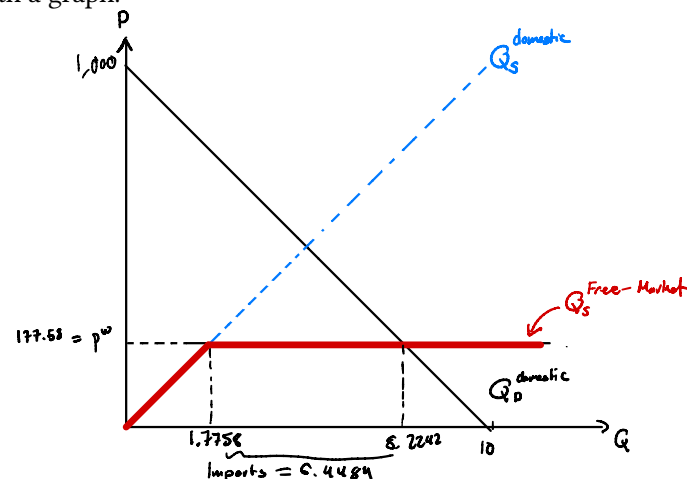
$$Q_D^{\text{domestic}} = 10 - \frac{1}{100}P; \quad Q_S^{\text{domestic}} = \frac{P}{100},$$

where the quantities are in millions of liters and the price is in KZT/liter.

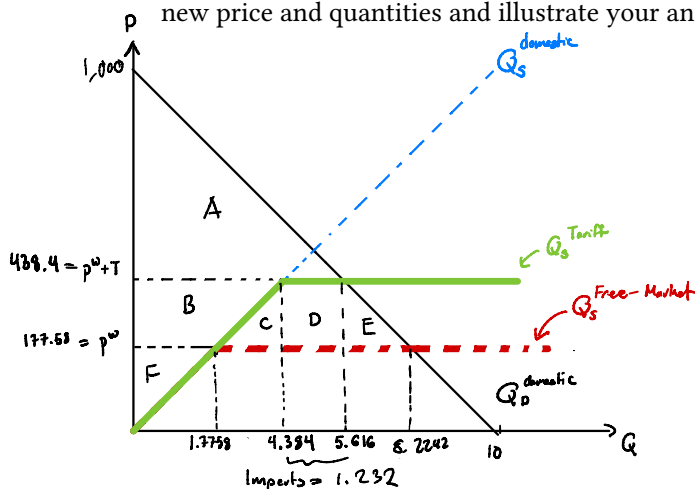
The world price of milk, in KZT/liter is currently $p^{\text{world}} = 177.58$.

- (a) (5 points) Consider first the situation of a *free-trade* market equilibrium. What would be the free-market equilibrium price and quantity? What would be the amount imported and the amount domestically produced? Calculate the items asked and also illustrate your answer with a graph.

At the Free Mkt. eq. $p^* = p^{\text{world}} = 177.58$
 $\Rightarrow Q^* = 10 - \frac{1}{100} 177.58 = 8.2242$ million of liters
 $\Rightarrow \text{Domestic Production} = \frac{177.58}{100} = 1.7758$ million of Liters
 $\Rightarrow \text{Imports} = 8.2242 - 1.7758 = 6.4484$



- (b) (5 points) In an effort to protect national milk producers, the government of Kazakhstan imposes a tariff of $T = 260.82$ KZT/liter. What would be the new market equilibrium after the tariff is imposed? Calculate the new price and quantities and illustrate your answer with a new graph.



$Q^* = Q_D^{\text{domestic}}(p^{\text{world}} + T) = 10 - \frac{1}{100}(438.4) = 5.616$
 The new eq. price is $p^{\text{world}} + T = 438.4$
 $\Rightarrow \text{Domestic production} = \frac{438.4}{100} = 4.384$
 $\Rightarrow \text{Imports} = 5.616 - 4.384 = 1.232$

- (c) (10 points) Fill the table (just input the numbers, you can do your calculations below):

	Free Trade	Tariff	Change
Consumer Surplus	A+B+C+D+E 3381.87	A 1576.58	-B-C-D-E -1805.29
Domestic Producer Surplus	F 157.673	B+F 960.923	B 803.3
Government Expenditures	0	D 321.33	D 321.33
Social Surplus	A+B+C+D+E+F 3539.55	A+B+D+F 2858.88	-C-E -680.664